The Assessment Question Paper and Assessment Answer Booklet must not be removed from the assessment room. Both must be handed in at the end of the assessment.

SIM Global Education

WRITTEN TEST
CONTINOUS ASSESSMENT (MAIN)

(DIPLOMA IN INFORMATION TECHNOLOGY)

Date	**:**	**22 April 2025**
Module	**:**	**ITSD001 Business Statistics with Python**
Time	**:**	**10:00 – 11:30**
Duration	**:**	**1 hour 30 minutes**
Total no. of pages (including this cover page)	**:**	**4**

INSTRUCTIONS TO CANDIDATES:

1. This assessment carries a total of 100 marks. Answer **ALL** questions.
2. Answer **ALL** questions in the answer booklet provided. Fill in the cover page carefully. **You will receive a zero (0) score if your name is filled incorrectly**.
3. All answers must be clearly written. Begin each question on a fresh page.
4. The use of non-programmable calculators **IS** permitted in this assessment.
5. This is a closed-book assessment. No unauthorised materials are allowed in the assessment venue.

At the end of the assessment:
Please ensure that you have written your name on each answer booklet used: failure to do so will mean that your work cannot be identified. If you have used more than one answer book, please tie them together with the string provided

SIM GLOBAL EDUCATION RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.

SAMPLE

ANSWER ALL QUESTIONS

Question 1

(a) Mr Lim is contemplating buying a small pre-owned car which is over 5 years old. After doing online research, he shortlisted the prices (in $'000) for ten such cars as shown below:

15	21	21	28	32	43	45	50	56	68

Compute the mean, median, mode, range and standard deviation. (Express your answer to 2 decimal places).

(10 marks)

(b) James is planning to start a new business selling souvenirs at Orchard Road. He carried out a simple survey to find out the spending on souvenir by tourists. Data were gathered from a sample of 10 tourists at one souvenir shop at Far East Plaza.

Amount Spent ($)	15	37	43	50	51	66	73	78	82	101

Construct a stem and leaf diagram from the above data. Use stem unit = $10 and leaf unit = $1.

(6 marks)

(c) Mr Tan decided to examine the daily sales of new cars by Lee Motors over the last 20 days:

Number of cars sold	**Number of days (Frequency)**	**Relative Frequency (%)**	**Class Mid-point**
2 up to 5	5		
5 up to 8	6		
8 up to 11	4		
11 up to 14	3		
14 up to 17	2		
Total	20		

Based on the frequency distribution table above, calculate the relative frequencies (Express your answer to 2 decimal places) and class mid-points. Construct a histogram based on the frequency distribution table provided. Label all axes clearly.

(9 marks)

SAMPLE

Question 2

(a) Mr Lim has decided to recruit 10 new employees and there were 80 applicants for the jobs. He has tabulated the 80 applicants into the following contingency table based on their age group and degree level.

Age Group	**Bachelor Degree Holders**	**Master Degree Holders**	**Total**
20 to 29	11	6	17
30 to 39	19	8	27
40 to 49	17	10	27
50 and above	8	1	9
Total	55	25	80

Given that a job applicant is from the age group 30 to 39, what is the probability that the person is a master degree holder? What is the probability that a job applicant belongs to the age group 40 to 49 or the applicant is a bachelor degree holder? (Express answer to 3 decimal places)

(8 marks)

(b) Given a normal distribution with mean $\boldsymbol{\mu}$= 30 and standard deviation $\boldsymbol{\sigma}$ = 5, what is P(30 < X < 34)? What is the probability a randomly selected sample of 25 customers has an average spending more than $32? (Express answer to 3 decimal place)

(8 marks)

(c) In a jackpot game, player can spend $80 on an opportunity to push a button to control the displaying window. If the window shows cherry, they get $200. If the window shows lemon, they get $300. If the window shows coin, they get $500. If the window shows others, they get nothing. The probability of getting cherry is 1/8, the probability of getting lemon is 1/27. The probability of getting coin is 1/64. What amount can a player expect to win or lose by playing the game?

(9 marks)

SAMPLE

Question 3

(a) A shop sells a particular layer cake. It is assumed that the daily sales of this type of cake is a Poisson variable with mean 5. Find the probability that on a particular day the shop sells:

(i) Exactly 6 cakes.
(ii) At least 2 cakes. (Correct your answers to 4 decimal places)

(10 marks)

(b) In a factory, on average, every one out of six earpieces is found to be defective. Twelve earpieces are selected at random. Find the probability that less than three of them will be defective. (Correct your answer to 4 decimal places)

(9 marks)

(c) State the level of measurement (Ratio, Ordinal or Nominal) for the variables: "Time spent revising for common test" and "Handphone number".

(6 marks)

Question 4

(a) A survey of 20 teachers in a school found that the mean age of the teachers is 40.6 years old, with a sample standard deviation of 9.5 years. Find the 99% confident interval of the population mean. (Express your answer to 3 decimal places)

(9 marks)

(b) An additional 20 teachers were surveyed. The mean age remained at 40.6 years and the sample standard deviation remained at 9.5 years. Find the 95% confident interval for the population mean age. (Express your answer to 3 decimal places.)

(9 marks)

(c) The surveyor has made a mistake in calculating the standard deviation and the correct sample standard deviation has a smaller value of 7. How would this affect the width of the confidence interval?

(7 marks)

--- END OF PAPER ----

SAMPLE